The Emotional Environment of Finn the Wolfhound: A Computational Validation of the 2002 Literary Theory

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Introduction

While reading the chapter "Dog Years, Human Fears" from Representing Animals, a question came up: do we give human traits to our favorite companions? I recently wrote a story about my own furry friends, Kobe and Sasha. I gave them human emotions, connecting the feelings of a human to a dog, and expressed those feelings toward the people I care about in the form of poetry. Analyzing the feelings of dogs is an interesting subject; dogs certainly do have feelings to some extent, even if they are different from human emotions. Teresa Mangum's 2002 essay raised a fascinating hypothesis about this exact dynamic. She argued that people in the 19th century often put their fictional pets in dangerous situations.The author had strong opinions on how writers used these environments to project human anxieties like old age onto animals.not knwing the plot of the book or reasing the story I first used a word cloud and led me to see the most common word was master. I decided to take a data driven aprroch being isolated in the wild where the dogs would theoretically experience intense fear is upto debate since their answestors were wolfs and should thrive instead of fear. This project computationally tests that exact idea using the text of Finn the Wolfhound. By analyzing the emotional vocabulary of the book, I wanted to see if the author mathematically manipulated the dog's feelings, shifting between "Trust" and "Fear" depending on whether Finn was safe at home or out in the wild. I will be using tools and algorithms to support or reject Magnums claims.

Objectives

The primary goal of this project is to use Natural Language Processing to mathematically map the emotional vocabulary of Finn the Wolfhound. Because dogs are descendants of wolves, they are naturally equipped to survive in the wild. I want to use statistical algorithms to see if the author artificially manipulated the narrative, forcing a spike in "Fear" words when Finn is in the wild and "Trust" words when he is with his master. Ultimately, this data will prove or reject Teresa Mangum's theory that 19th-century authors intentionally engineered these environments to mirror human anxieties about aging and isolation.

Methods

Methods To computationally test the 2002 hypothesis, I built a Natural Language Processing (NLP) and statistical pipeline using R:

Data Preprocessing: I retrieved the text from Project Gutenberg and used Regular Expressions (Regex) to extract and align the chronological chapter titles. I then used Tokenization to chop the text into individual words and filtered out standard English stop words. Word count analys: checked most common words to make a decision of the most useful data. "master" resebled and connects to perfectly witht he dieas of theresa where she also gave acklogement to the word.

Algorithmic Stemming: I applied the Snowball Stemming Algorithm (SnowballC) to group word variations together. Lexicon-Based Sentiment Analysis: I used the NRC Emotion Lexicon to map emotional values to the cleaned text. I specifically isolated and measured the frequencies of "Trust"and "Fear".

Unsupervised Machine Learning: I initially ran Clustering algorithms to search for emotional timelines. However, I abandoned this clustering approach when the algorithms identified extreme outlier chapters rather than the narrative environments.

Statistical Inference: To establish the environments, I used a Median Split on the frequency of the word "master" to divide the book into "Domestic" and "Wild" halves. Finally, I executed a Pearson's Chi-Squared Test with Yates' Continuity Correction to mathematically prove whether the shift in "Trust" and "Fear" across these environments was statistically significant.

Results

while takling this anis i noticed i do not know the plot of the book.I undertand my internal biased of wanting dogs to be placed in a truting environment. my aproced and decisions were driven by We confidently reject the null hypothesis. The emotional tone of Finn the Wolfhound is mathematically dependent on the dog's setting. The data reveals a deeper narrative layer: the emotion of "Trust" in the novel is entirely tethered to the dog's "master worship." In the chapters where Finn is isolated in the wild, this master worship is physically missing from the text, causing "Trust" vocabulary to completely plummet. Without the master present to anchor the dog's trust, the author deliberately surges "Fear" vocabulary to fill the void and maximize the narrative drama of the wild. This provides robust, data-driven validation for Teresa Mangum's 2002 literary theory.

Next Steps

teresa raised another question on wether people used human charactericstics mostly in dogs. She raised the questions if books about horses had any different sentiments. I woould like to explore this since it connects to human behaviours.

Conclusion

References